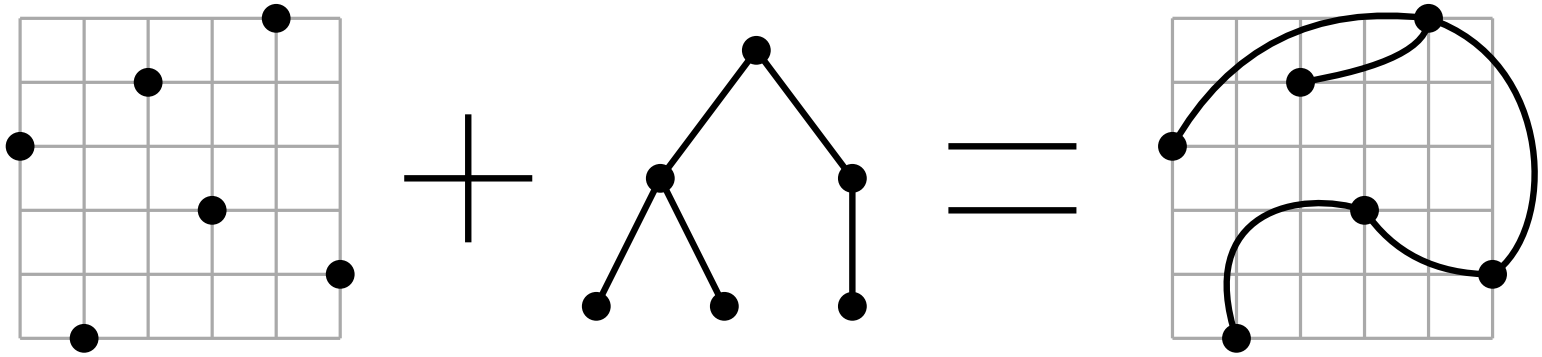
Planar L-Shaped Point Set Embeddings of Trees

O. Aichholzer, T. Hackl, M. Scheucher

Point Set Embeddings

T ... tree on n vertices
P ... set of m points

point set embedding ... drawing of T, vertices drawn as points of P



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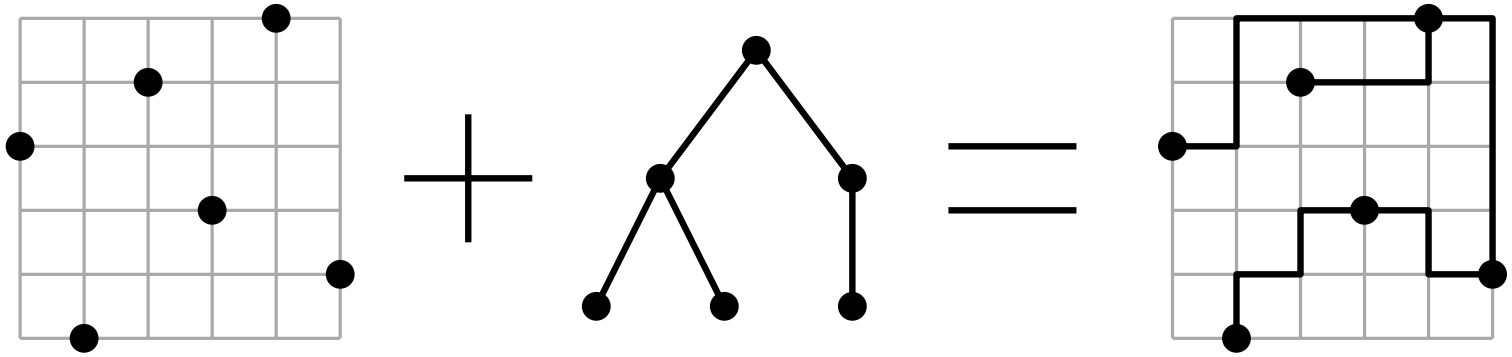

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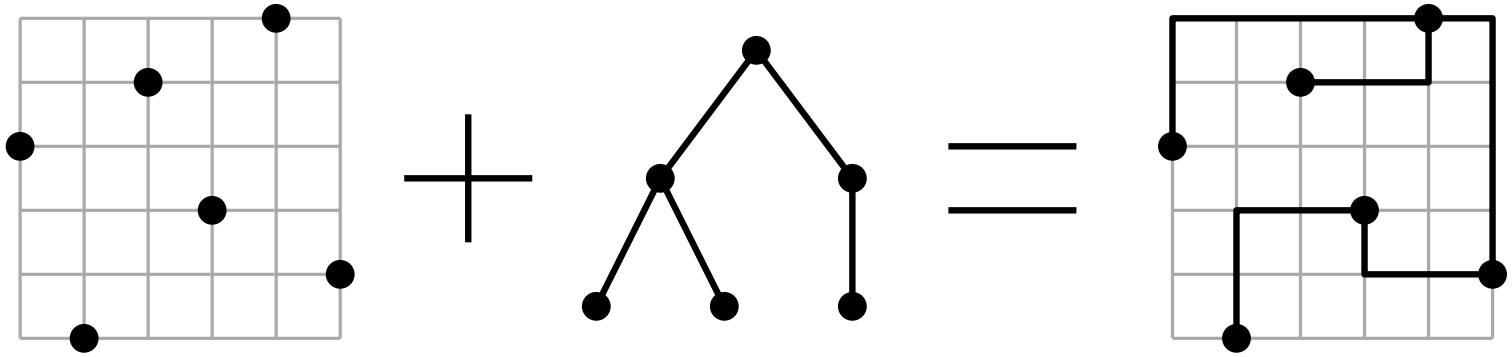

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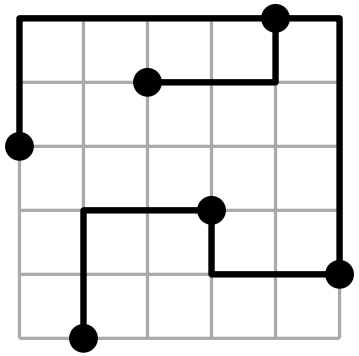

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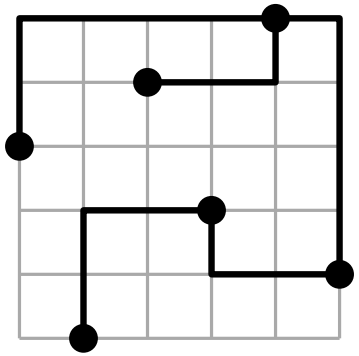
- distinct x- and y-coordinates
- $P = \{(1, \pi_1), \ldots, (m, \pi_m)\}$
  (π ... permutation)



Point Set Embeddings

$f(T)$ ... minimum number m : tree T admits a planar L-shaped embedding in any set of m points

$$f_d(n) := \max_{\substack{T \text{ : tree on } n \text{ vertices} \\ \text{max. deg. } \Delta(T) \leq d}} f(T)$$

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	Planar L-Shaped	Nonpl. L-Shaped	Planar Orthogeod.
Cat. $\Delta = 3$	n	n	n
Tree $\Delta = 3$	$n^2 - 2n + 2$	n	$1.5n$
Cat. $\Delta = 4$	$3n - 2$	$n + 1$	$n + 1$
Tree $\Delta = 4$	$n^2 - 2n + 2$	$4n - 3$	$4n - 3$

Table: Upper bounds on the number of required points by Di Giacomo, Frati, Fulek, Grilli, and Krug (2013)

Subquadratic Upper Bound

Theorem: $f_4(n) \leq n^{\log_2 3} \qquad (\log_2 3 \approx 1.585)$

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Proof outline:

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- $u_n \leq n^{\log_2 3}$
- T ... tree on n vertices ($\Delta \leq 4$)
  P ... set of u_n points
  Recursive approach:
  find planar L-shaped embedding of T in P
  Recursive step:
  embed subtrees of sizes a, b, c
  in sub point sets of sizes u_a, u_b, u_c, resp.

Proof of the Lemma (1 of 2)

Lemma: $u_n \leq n^{\log_2 3}$

Proof: $g(x) := x^{\log_2 3}$ on $[0, \infty)$

Base: $u_0 = 0 = g(0)$, $u_1 = 1 = g(1)$

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Proof: $g(x) := x^{\log_2 3}$ on $[0, \infty)$

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Step: Assume $u_k \leq g(k)$ for $k \leq n$

$S := \{(x, y, z) \in \mathbb{R}^3 \colon x \geq y \geq z \geq 0,\ x + y + z = n\}$.

As $u_n = 1 + u_a + 2u_b + 2u_c$ for some $a + b + c = n$,

$$\begin{aligned} u_{n+1} &\overset{IA}{\leq} 1 + g(a) + 2g(b) + 2g(c) \\ &\leq \max_{(x,y,z) \in S} \underbrace{1 + g(x) + 2g(y) + 2g(z)}_{=:h(x,y,z) \text{ (convex)}} \end{aligned}$$

Proof of the Lemma (2 of 2)

h ... convex function

S ... convex set, spanned by

$$s_1 = (n, 0, 0),\ s_2 = (\tfrac{n}{2}, \tfrac{n}{2}, 0),\ s_3 = (\tfrac{n}{3}, \tfrac{n}{3}, \tfrac{n}{3})$$

$\overset{\text{Maximum Principle}}{\Longrightarrow}$ h attains maximum (over S) in s_1, s_2, or s_3

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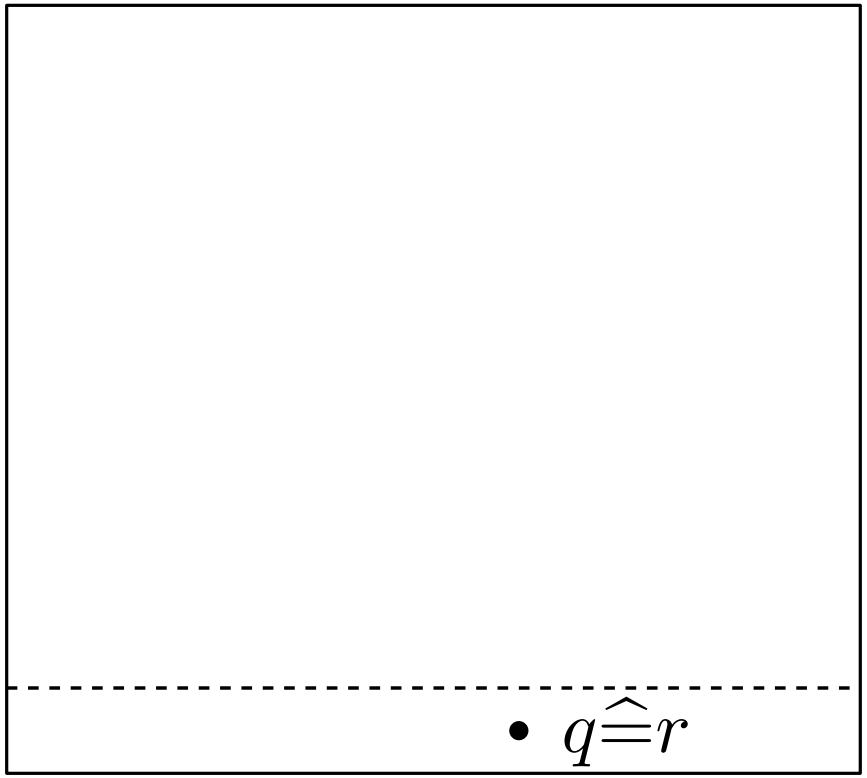
- $h(s_1) = 1 + g(n) \leq g(n+1)$
- $h(s_2) = 1 + 3(n/2)^{\log_2 3} = 1 + n^{\log_2 3} = 1 + g(n) \leq g(n+1)$
- $h(s_3) = 1 + 5(n/3)^{\log_2 3} \leq 1 + 1 \cdot n^{\log_2 3} \leq 1 + g(n) \leq g(n+1)$

$\Longrightarrow u_{n+1} \leq g(n+1)$ $\square$

Recursive Embedding Approach

T ... tree on n vertices ($\Delta \leq 4$) rooted at vertex r of deg. 1
P ... set of u_n points

Place r at bottommost point q of P



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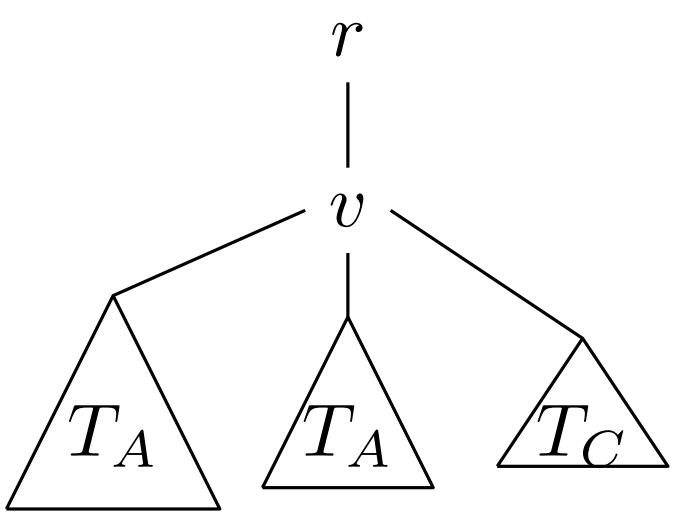
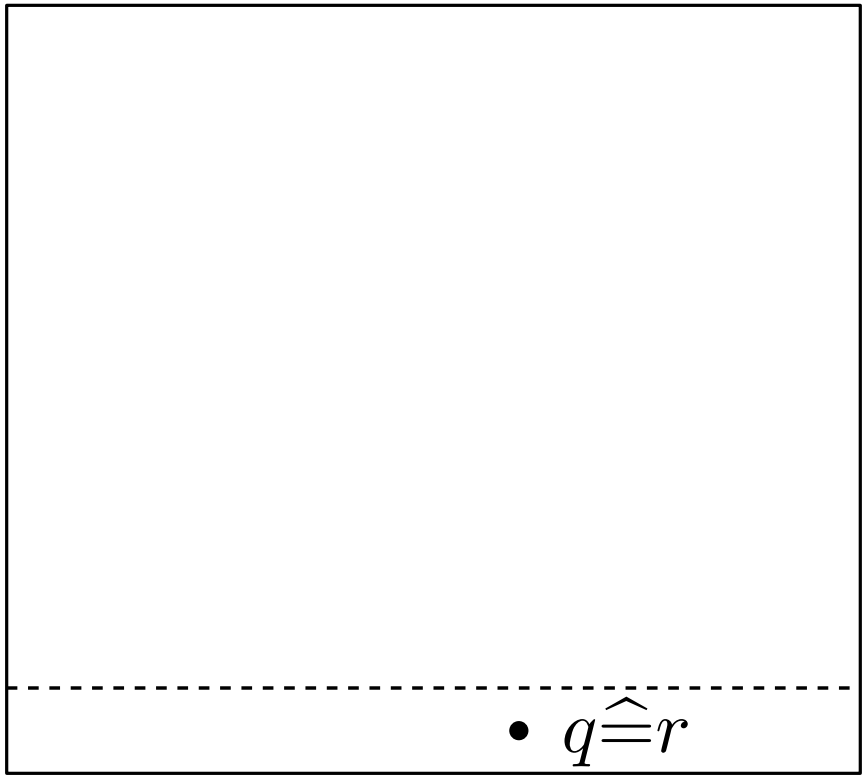
Child v of r: subtrees
T_A, T_B, T_C of sizes $a \geq b \geq c$
$(n - 1 = 1 + a + b + c)$





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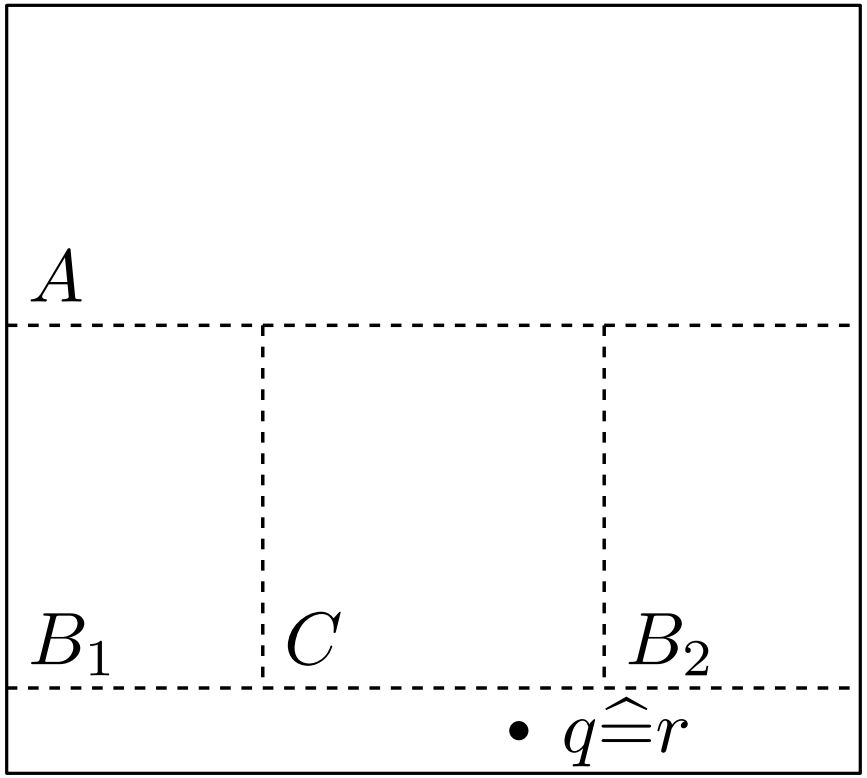

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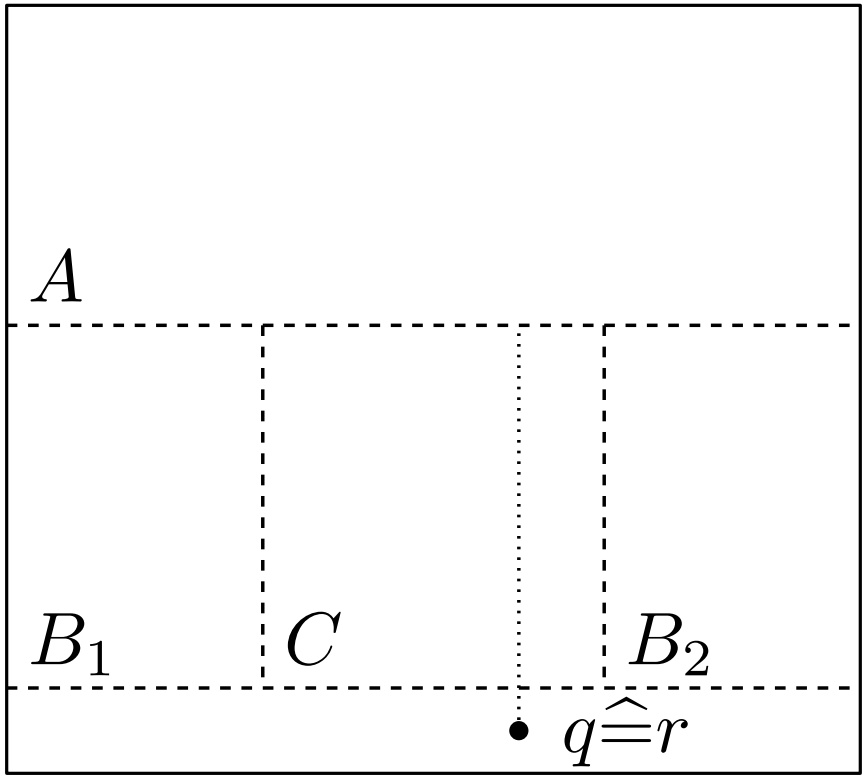

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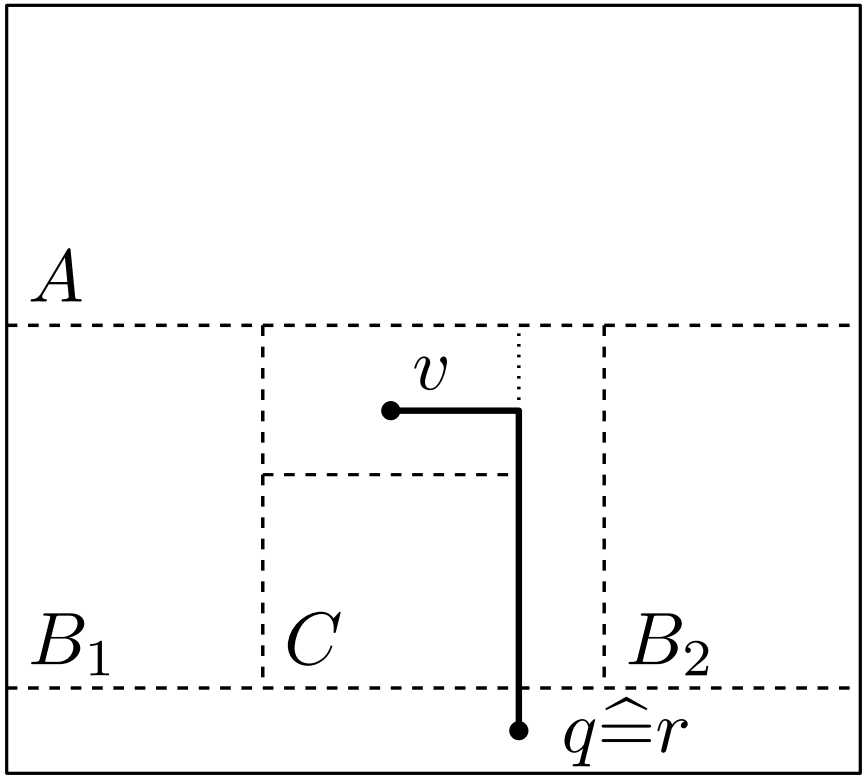

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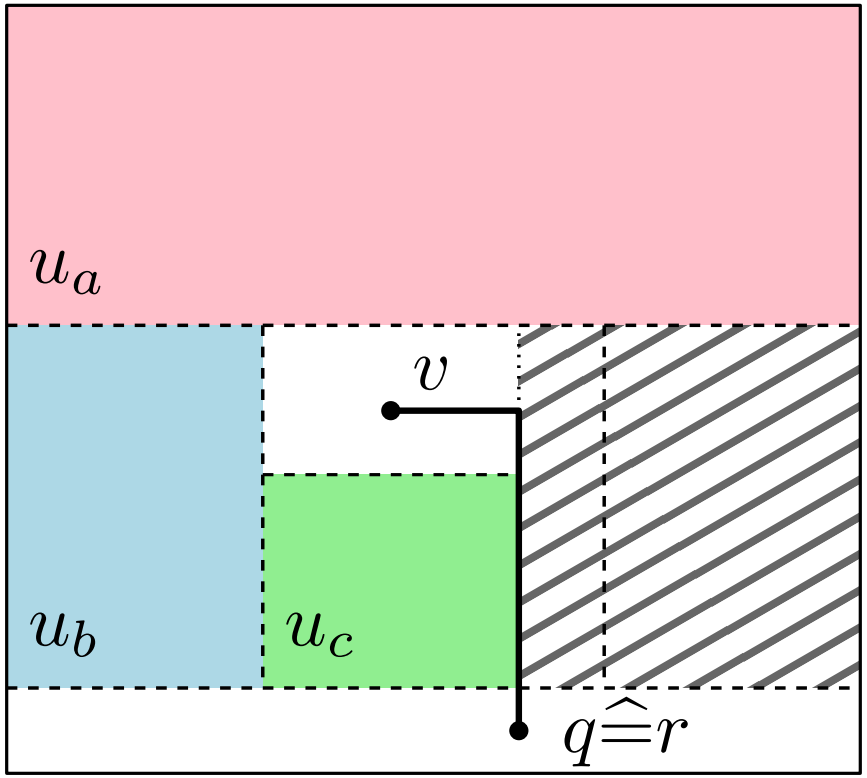

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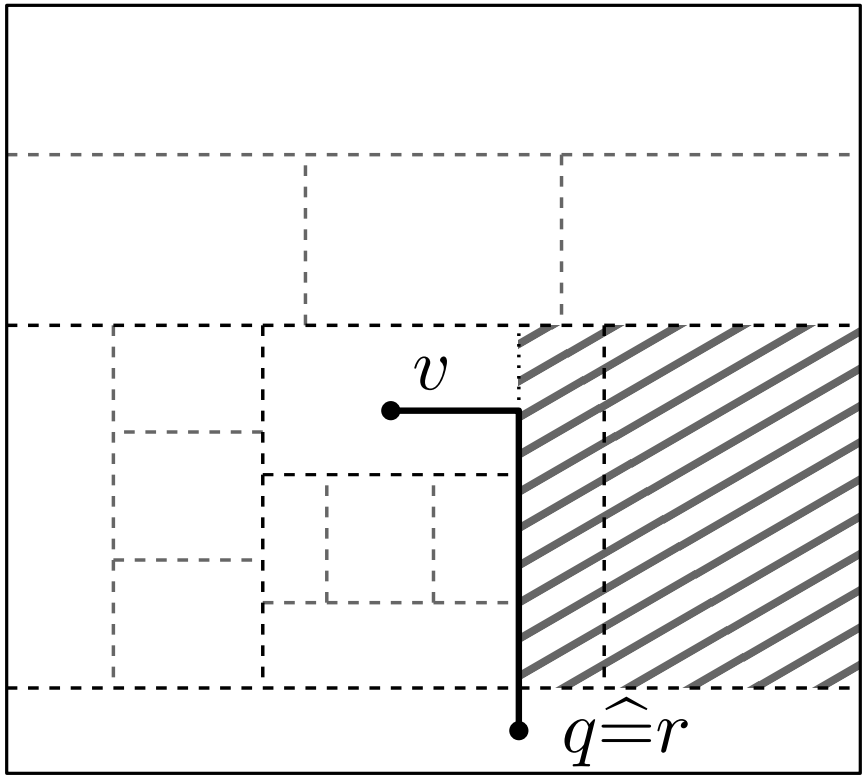

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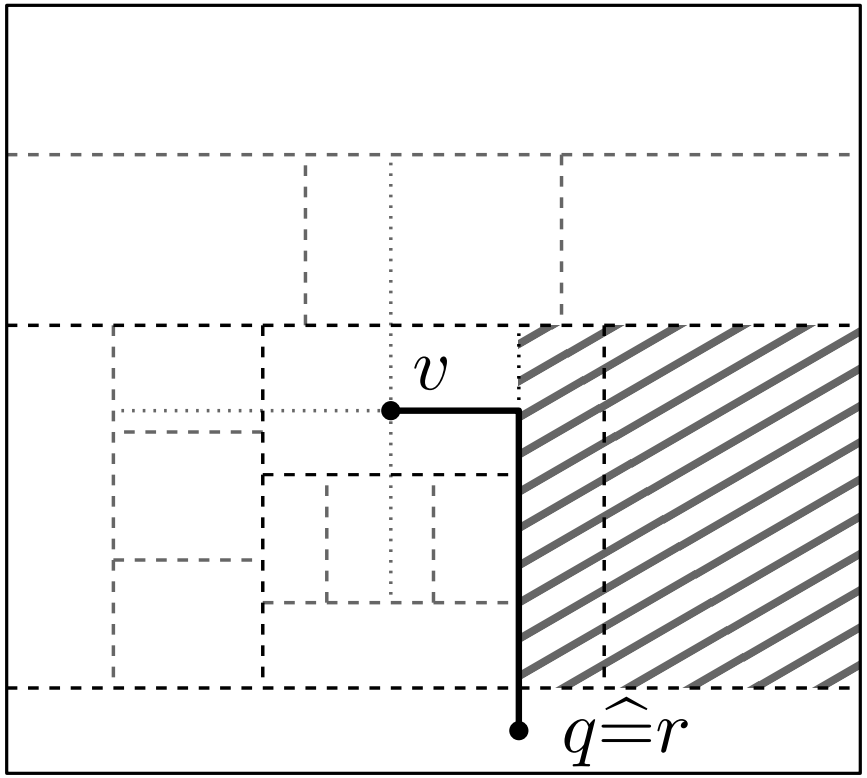

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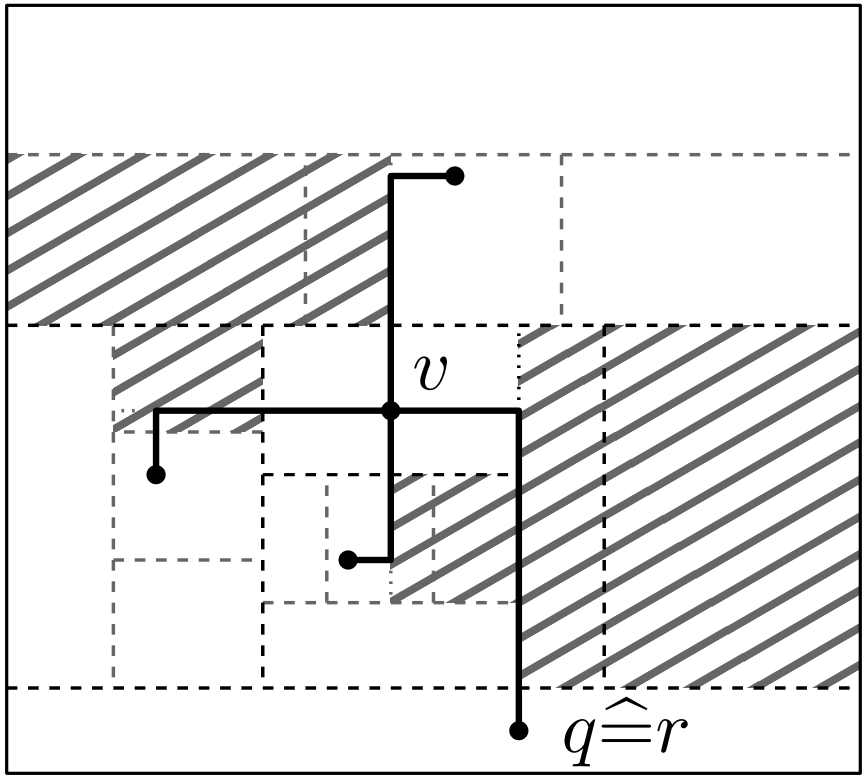

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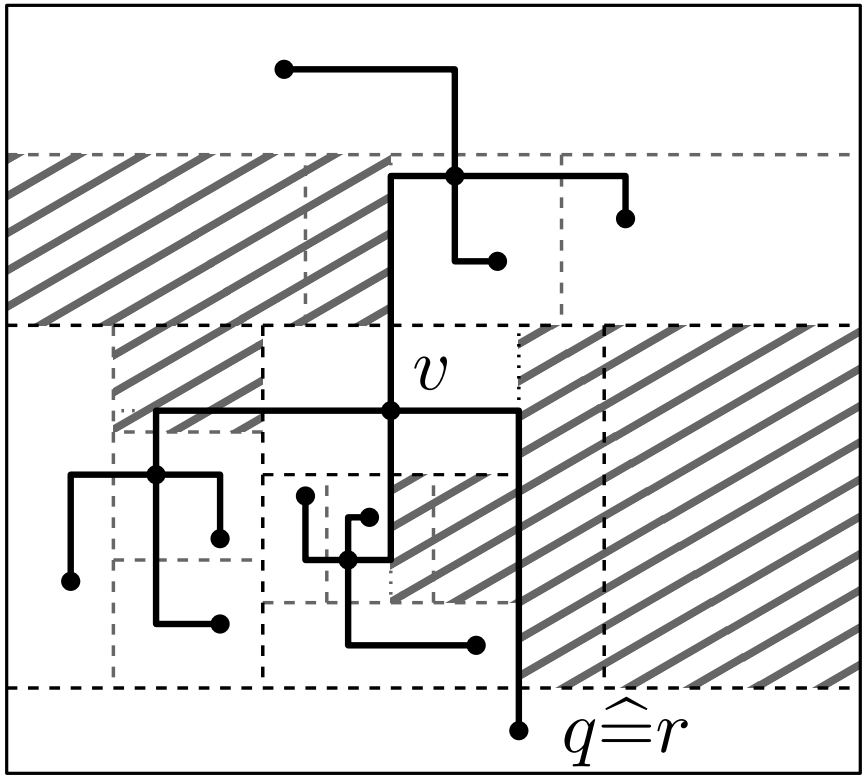

Subquadratic Upper Bound

Theorem: $f_4(n) \leq n^{\log_2 3} \quad (\log_2 3 \approx 1.585)$

Previous Best Bound: $f_4(n) \leq n^2 - 2n + 2$
[Di Giacomo et al., 2013]

Further Improvements:

- $f_3(n) \leq 0.5n^{\log_2 3} + O(n)$
- $f_4(n) \leq cn^{\log_2 3} + O(n)$, $c \approx 0.508$

Special Cases

T ... tree with vertex set V and root r

$$\sigma_r(v) := \max\{0, \sigma_r(u_1), \sigma_r(u_2) + 1, \ldots, \sigma_r(u_k) + 1\}$$

$u_1, \ldots, u_k$ children of v,

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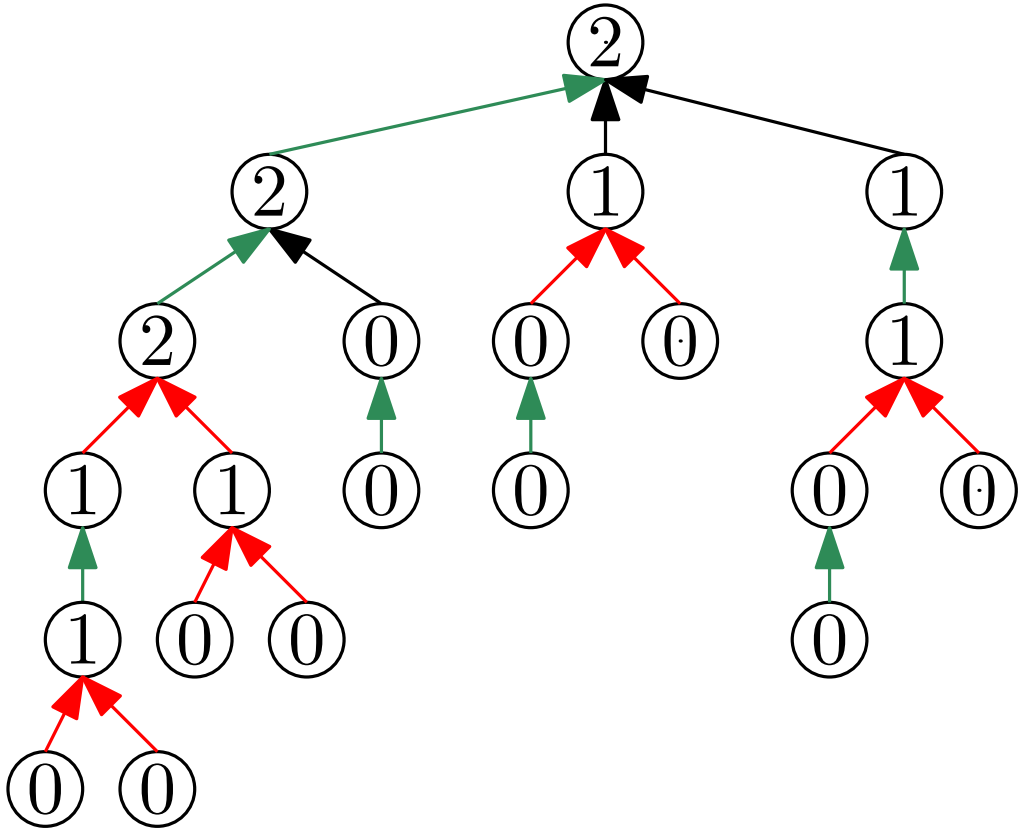

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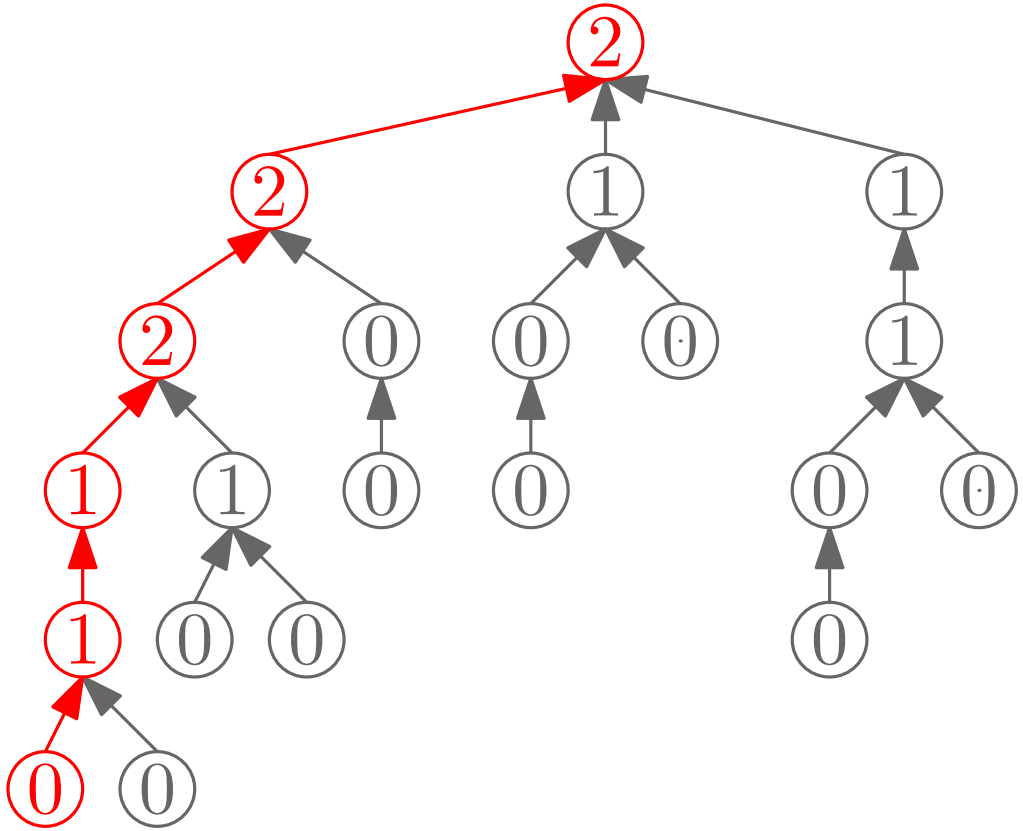

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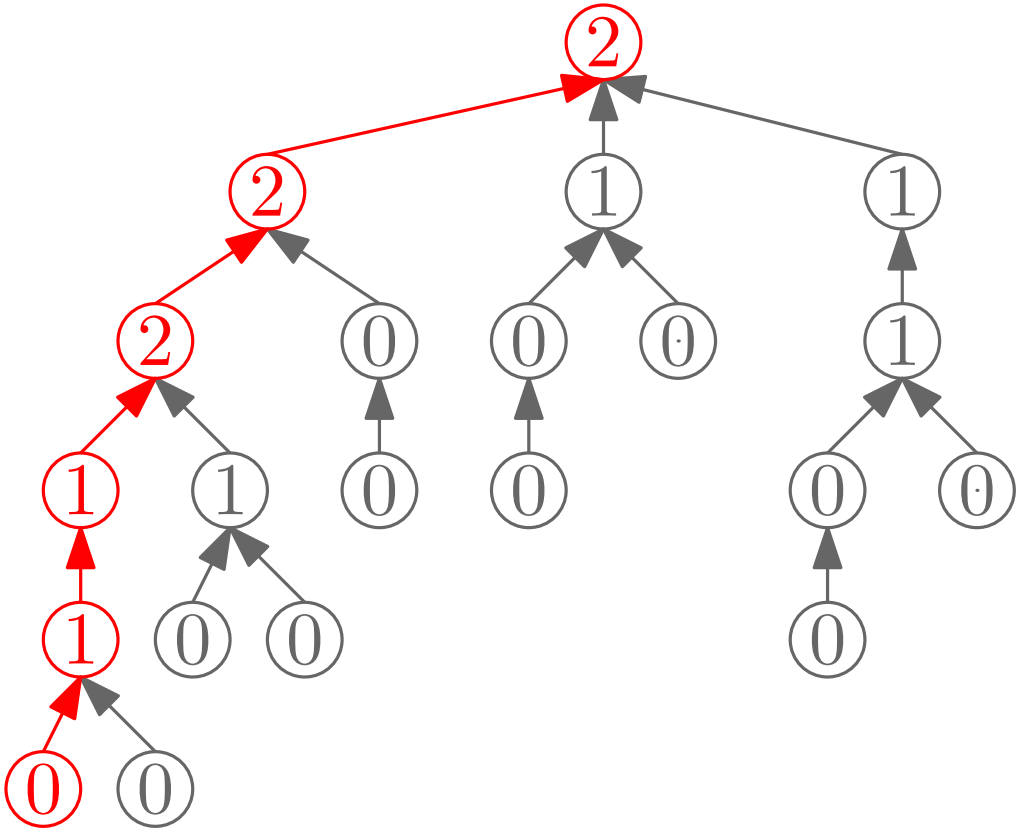

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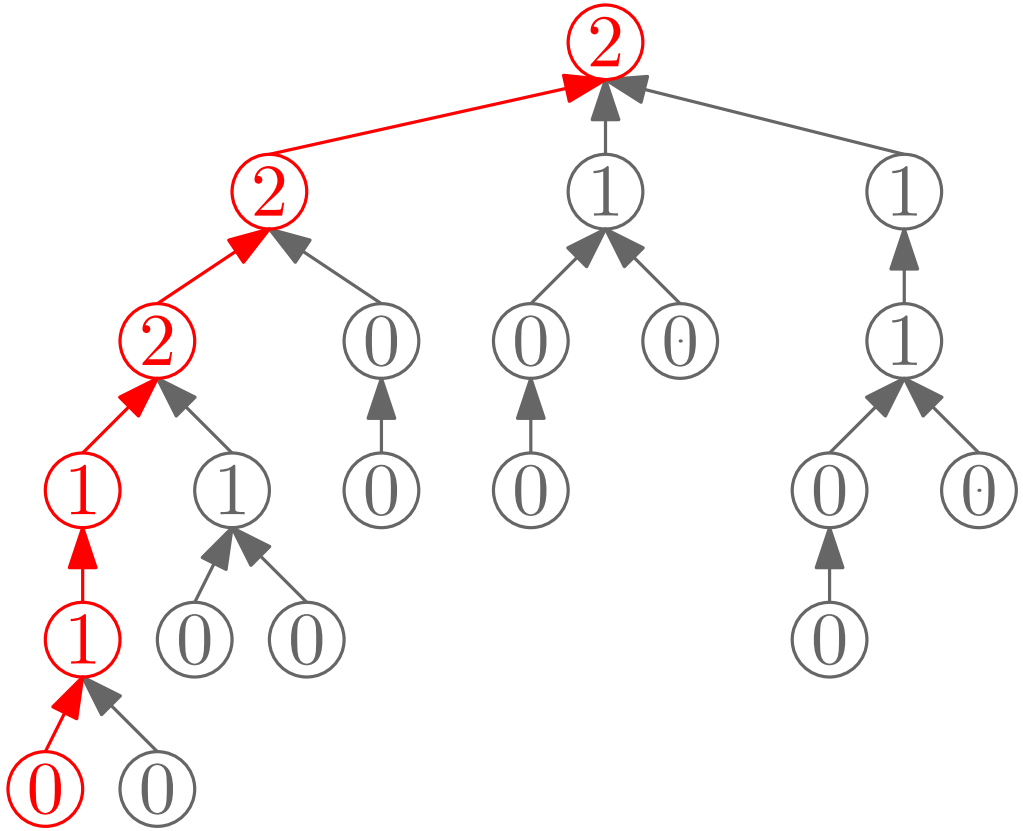
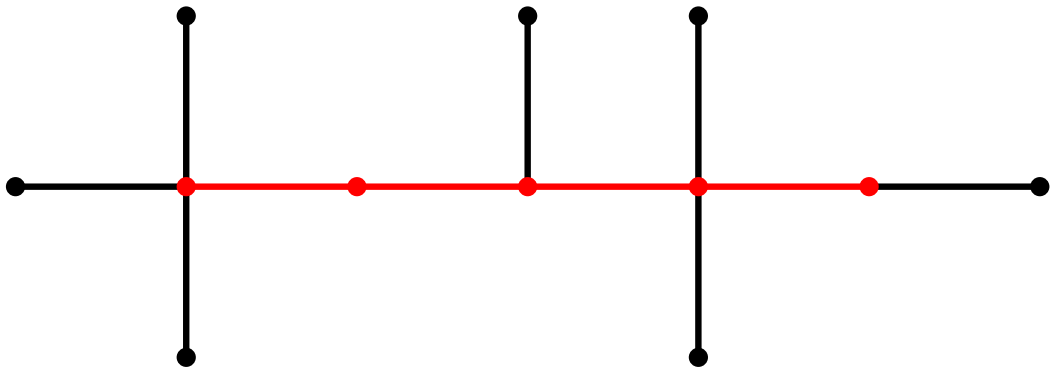
Theorem: $f(T) \leq 2^{\sigma(T)} n$ holds for every tree T with n vertices and $\Delta(T) \leq 4$



Caterpillars

caterpillar . . . tree : removal of leaves results in a path



Corollary: $f(T) \leq 2n$ holds for every caterpillar T with n vertices and $\Delta(T) \leq 4$.

Previous Best Bound for Caterpillars: $f(T) \leq 3n - 2$ [Di Giacomo et al., 2013]

Further Improvements: $f(T) \leq (4/3 + \varepsilon)n$ holds for every caterpillar T with n vertices and $\Delta(T) \leq 4$.

(Rooted) Trees Chosen Uniformly at Random

T_n ... set of rooted trees with n vertices and $\Delta \leq 4$

Consider T_n as random variable
(every tree in T_n is equally likely)

Theorem: Let $\varepsilon > 0$. Then the probability $\mathbb{P}[f(T_n) = O(n^{1.5+\varepsilon})]$ is at least $p = \frac{2\varepsilon}{1+2\varepsilon} > 0$

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Proof Outline:

- *register function* ρ (upper bound of σ)

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Theorem: Let $\varepsilon > 0$. Then the probability $\mathbb{P}[f(T_n) = O(n^{1.5+\varepsilon})]$ is at least $p = \frac{2\varepsilon}{1+2\varepsilon} > 0$

Proof Outline:

- *register function* ρ (upper bound of σ) has expected value $\mathbb{E}(\rho(T_n)) = \log_4 n + O(1)$ [Drmota and Prodinger, 2006]
- Follows from $f(T) \leq 2^{\sigma(T)} n$ and Markov's inequality

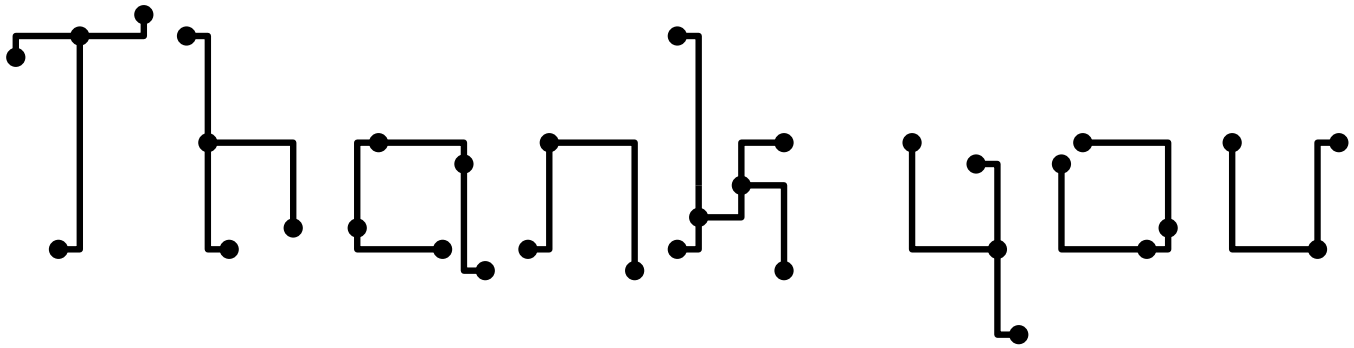
Point Sets Chosen Uniformly at Random

$f^{1/2}(T)$ ... minimum number m : tree T admits a planar L-shaped embedding in at least half of all sets of m points

$$f_d^{1/2}(n) := \max_{\substack{T \text{ : tree on } n \text{ vertices} \\ \text{max. deg. } \Delta(T) \leq d}} f^{1/2}(T)$$

Theorem: $f_3^{1/2}(n) = O(n \log n (\log \log n)^2)$

Theorem: $f_4^{1/2}(n) = O(n^{c+\varepsilon})$ for $\varepsilon > 0$, $c \approx 1.332$



- $f_4(n) = O(n^{\log_2 3})$
- $f(T) \leq 2^{\sigma(T)} n$
- $f_3^{1/2}(n) = O(n \log n (\log \log n)^2)$